Networks & Distributed Programming

The Application Layer: SMTP

Lecture 6



UNIVERSITÀ DEGLI STUDI DI NAPOLI
FEDERICO II



DIETI

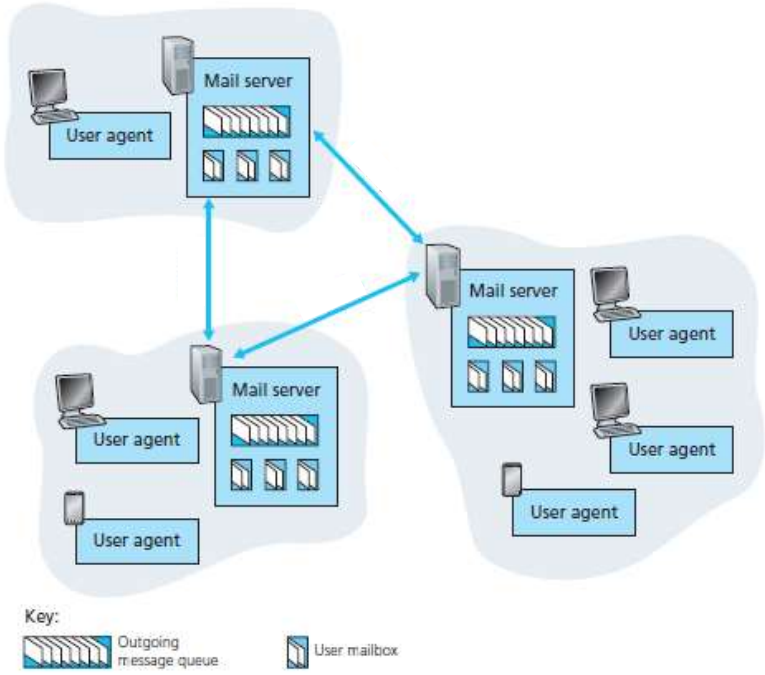
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Application Layer
Mails and SMTP

- Electronic mailing (e-mail) is one of the oldest and most important internet applications.

- Mailing is typically implemented in a client-server fashion, having 2 types of hosts:
  - **User agent**: application allowing mail management (e.g., Pine, K-9 Mail etc.).
  - **Mail Server**: a server that stores mails and maintain user-specific mailboxes.

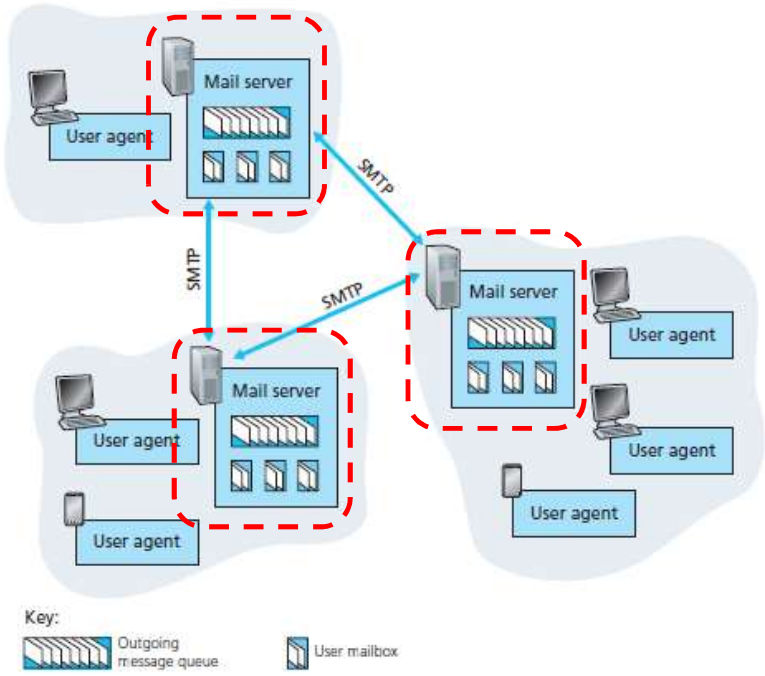
- Differently from HTTP, here **communication also occurs between servers**. We need 2 applications/protocols:
  - One **between user agent and mail server**.
  - One **between mail server and mail server**.



Application Layer
Mails and SMTP

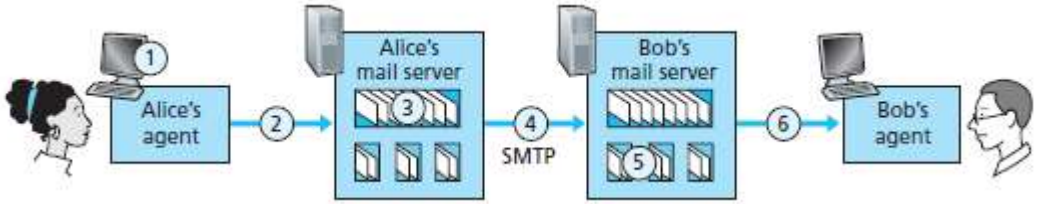
- Users **do not exchange mails directly** (as in instant messaging). It is preferred to use the more reliable and specialized mail servers.

- The **Simple Mail Transfer Protocol** (SMTP) is the main application-layer protocol working between **mail servers**.
  - It allows servers to exchange mails, **user agents do not use SMTP**.
  - Once a new mail arrives on the server, it is stored inside the **user-specific mailboxes**, waiting to be download locally by the user agent.
  - In SMTP there are a **client-side** (sender) and a **server-side** (receiver) that run on mail servers, both using the reliable **TCP** to transfer mail.



Application Layer
SMTP

- Let's assume to have host A (Alice) that is sending an e-mail to host B (Bob), we have 4 elements involved:
  - Alice's agent.
  - Alice's mail server.
  - Bob's agent.
  - Bob's mail server.



- The process works as follows:
  1. Alice **invokes her user agent** for e-mail, provides Bob's e-mail address (e.g., bob@someschool.edu), composes a message, and instructs the user agent to send the message.
  2. Alice's **user agent sends the message to her mail server**, where it is placed in a message queue.
  3. The client side of SMTP, running on Alice's mail server, sees the message in the message queue. It **opens a TCP connection to an SMTP server**, running on Bob's mail server.
  4. After some initial SMTP handshaking, the SMTP client sends Alice's message **into the TCP** connection.
  5. At Bob's mail server, **the server side of SMTP receives the message**. Bob's mail server then places the message in **Bob's mailbox**.
  6. Bob **invokes his user agent** to read the message at his convenience.

Application Layer
SMTP

- An SMTP dialogue

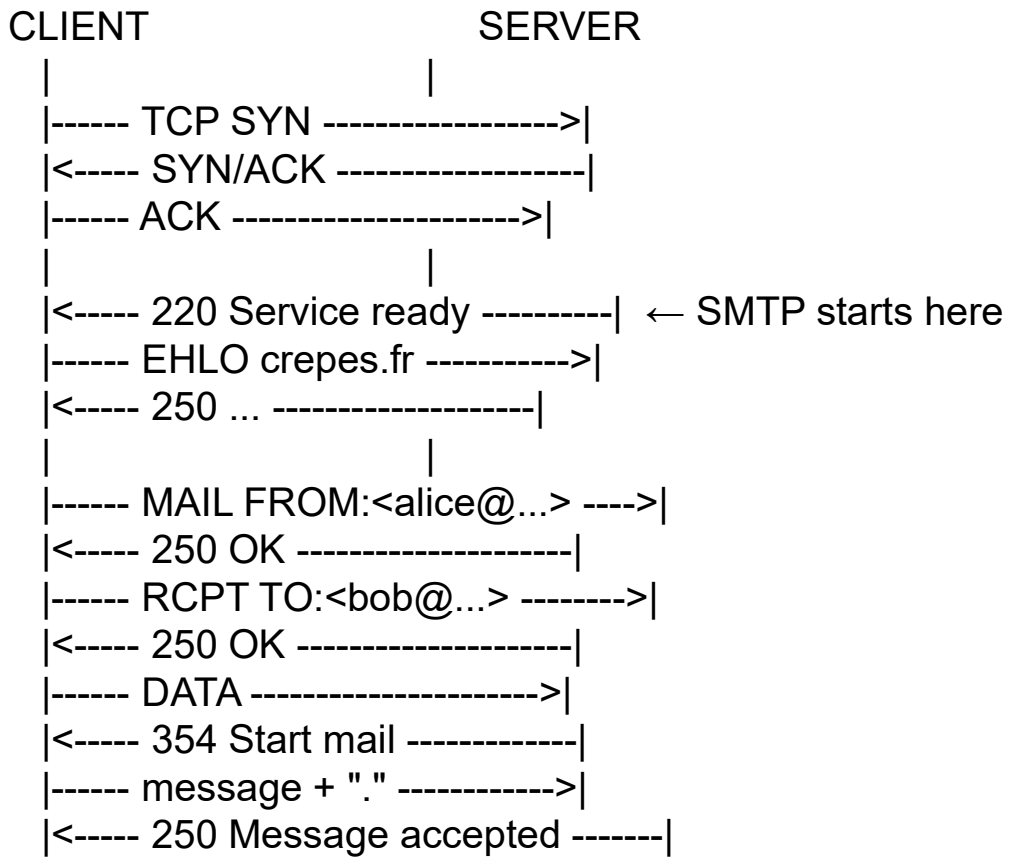
Client `crepes.fr`, server `hamburger.edu`, TCP already established.

```
S: 220 hamburger.edu
C: HELO crepes.fr                    today EHLO: the extended greeting
S: 250 Hello crepes.fr, pleased to meet you
C: MAIL FROM: <alice@crepes.fr>
S: 250 alice@crepes.fr ... Sender ok
C: RCPT TO: <bob@hamburger.edu>
S: 250 bob@hamburger.edu ... Recipient ok
C: DATA
S: 354 Enter mail, end with "." on a line by itself
C: Do you like ketchup?
C: .                                 a lone period ends the message
S: 250 Message accepted for delivery
C: QUIT
```

Five commands, and a numeric reply to each one — the same idea as an HTTP status code. The connection is persistent: more messages follow here.

Application Layer
SMTP

- An SMTP dialogue



Application Layer
SMTP

- The message

Inside `DATA` travels a message with a header of its own, defined by a different standard:

```
From: alice@crepes.fr
To: bob@hamburger.edu
Subject: Searching for the meaning of life.

(the body, after the blank line)
```

- These are not the SMTP commands, even where the words coincide: the commands drive the delivery, these travel inside it.
- SMTP is older than HTTP and restricts the whole message, body included, to 7-bit ASCII.
- So an image is encoded into letters before it leaves and decoded on arrival — which is why an attachment is bigger than the file.

Application Layer
SMTP

- First, the client SMTP (running on the sending mail server host) **establishes a TCP a connection to port 25 at the server SMTP** (running on the receiving mail server host).

- If the server is down, the client tries again later. Once this connection is established, **the server and client perform some application-layer handshaking**: SMTP clients and servers introduce themselves before transferring information.
  - During this SMTP handshaking phase, the SMTP client indicates the e-mail address of the sender (the person who generated the message) and the e-mail address of the receiver.

- SMTP can count on **the reliable data transfer service of TCP** to get the message to the server without errors.

Application Layer
Accessing mails

- Deployment of SMTP servers is **more reliable** with respect to direct user-to-user mailing:
  - Servers are, by definition, always on and visible to all devices of the network.
  - Servers are certified.
  - The server may continuously try to re-send mails if delivery fails (which is a quite computationally expensive process).

- On the other hand, to access the e-mails from the server **an additional client-server application is needed** (and a different protocol), most popular are:
  - Post Office Protocol—Version 3 (**POP3**).
  - Internet Mail Access Protocol (**IMAP**).
  - **HTTP**.

Application Layer

Accessing mails: POP3

- The **Post Office Protocol - Version 3** (POP3) is an old and extremely simple mail access protocol.

- The user agent (the client) opens a **TCP connection** to the mail server (the server) on port 110.

- With the TCP connection established, POP3 progresses through three phases:
  1. **Authorization**: the user agent **sends a username and a password** to authenticate the user.
  2. **Transaction**: the user agent can **retrieve messages**, mark messages for deletion, remove deletion marks, and obtain mail statistics.
  3. **Update**: after the client has issued the quit command, ending the POP3 session, the mail server **deletes the messages** that were marked for deletion.

Application Layer

Accessing mails: IMAP

- With POP3 access, messages can just be **downloaded** or **deleted**. Action like searching or organizing mails into folders are not considered (these must be done on the local machine).

- The **Internet Mail Access Protocol** (IMAP) allow servers to provide **additional features:**
  - Managing and **creating folders.**
  - Perform **search** into remote folders for messages matching specific criteria.
  - Allow user agent fetch parts of messages (e.g., header only, attachments only, etc.).
  - Allow **multiple clients** to be connected to the same server.

Application Layer

Accessing mails: HTTP

- More and more users today are sending and accessing their e-mails through **their Web browsers**.
- HTTP Web-based access was introduced by Hotmail in the mid 1990 an is now the mainstream approach (Google, Yahoo!, Virgilio, etc.) and it is used by almost every major university and corporation (UniNA included).
- With this service, **the user agent is an ordinary Web browser**, and the user communicates with its remote mailbox via HTTP rather than through the POP3 or IMAP protocols.
- This works for user-agents, while **mail servers still rely on the standard SMTP** to exchange messages.